

Combining PageRank Modularization and Ontology Verbalization for LLM-Based Complex Ontology Matching

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Abstract

Complex matching remains a challenge in the ontology matching field. Existing approaches are constrained by strong assumptions on their input: instance-based and rule-mining systems require populated ontologies and reliable links between their instances, while systems built on pre-trained encoders operate at the schema level but target primarily simple correspondences. Recent approaches based on Large Language Models (LLMs) are promising, yet they either rely on manually built ontology modules or feed formal serializations directly to the model. This paper proposes a schema-only pipeline that addresses these limitations by integrating (i) an automatic modularization strategy combining the PageRank and Personalized PageRank algorithms, which reduces the search space while controlling the specificity of the extracted context; and (ii) a verbalization module that turns ontology modules into natural language descriptions, exploited both for embedding-based candidate generation and as context in LLM prompts. The approach is evaluated on four datasets of the OAEI Complex track. Relying solely on schema-level information, our matcher achieves competitive performance compared with state-of-the-art complex matching systems.

Keywords

complex matching, modularisation, LLM, verbalization, PageRank

1. Introduction

Ontology matching is the task of identifying correspondences between semantically related entities belonging to different ontologies, thereby facilitating interoperability and knowledge integration. Traditionally, the majority of matching systems have focused on simple correspondences. However, real-world ontologies frequently exhibit rich heterogeneity in the way they model knowledge (in terms of granularity, relations, or structures) that cannot be always reconciled by simple (1:1) correspondences [1]. Complex correspondences (1:n, n:1, and n:n) provide a more accurate representation of knowledge, but their automatic discovery remains challenging. These correspondences are more expressive, since they make it possible to capture nuances such as property chains, class unions, or value transformations across ontologies [2]. For example, a complex correspondence can establish that a class in one ontology corresponds to a restriction or a combination of classes in another ontology. To illustrate, given two ontologies o_1 and o_2 , here are some examples:

$$o_1:AcceptedPaper \equiv o_2:Paper \sqcap \exists o_2:acceptedBy.\top$$

constitutes a *complex correspondence* involving a *logical constructor* (an existential property restriction combined with a class intersection, corresponding to the *Class by Attribute Existence* pattern);

$$o_1:authorOf \equiv (o_2:writtenBy)^-$$

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constitutes a *complex correspondence* involving a *logical constructor* (inverse property).

Existing approaches to complex ontology matching, however, remain constrained by strong assumptions on their input. Instance-based systems such as CANARD [3] require populated ontologies and reliable sameAs links between them, a requirement seldom satisfied in practice, and favour coverage at the expense of precision. Rule-mining approaches such as AROA [4] likewise depend on shared instances and are restricted to a predefined set of correspondence patterns, which limits their generalisation to unseen forms of heterogeneity. Systems built on pre-trained encoders, such as BERTMap [5], operate at the schema level, but were designed primarily for simple correspondences and rely on similarity between isolated labels, without the compositional reasoning required to assemble logical constructors. Conversely, generative Large Language Models (LLMs) applied to matching without structural constraints tend to produce an excess of false positives [6].

This paper proposes a novel approach for the generation of complex ontology matching, based on ontology *modularization* and *verbalization*. These two components address, in a complementary manner, the main obstacles that currently limit the application of LLMs to complex ontology alignment. On the one hand, real-world ontologies are too large to be provided in their entirety as input to an LLM: beyond the practical limits imposed by the context window, overly long inputs are known to degrade model performance, as relevant information gets diluted among irrelevant content. Modularization directly tackles this issue by reducing the search space: it extracts relevant sub-ontologies around candidate entities, while preserving the essential structural context and mitigating noise. On the other hand, ontology serializations (e.g., RDF/XML or Turtle) do not correspond to the natural language distribution on which LLMs are predominantly trained, which hinders their ability to reason over such representations. Verbalization [7] bridges this gap by translating formal structures (classes, properties, and axioms) into textual descriptions, allowing both embedding-based similarity methods and generative language models to capture fine-grained conceptual nuances that would remain inaccessible in purely symbolic form. Alignments thus become not only computationally tractable (through modularization) but also *semantically informed*, in the sense that correspondences are grounded in explicit textual descriptions of the meaning of entities rather than in surface-level lexical similarity.

The novelty of the approach lies in the integration of these two components into a unified LLM-based alignment pipeline, where they are jointly exploited at two distinct stages: during candidate generation, where verbalized modules are used to produce semantic embeddings for similarity-based filtering, and during the alignment phase itself, where they provide contextualized natural language input to the prompts of the LLM. This contrasts with previous work, in which modularization is performed manually [8], modules are provided to the model in raw Turtle syntax [1], or modularization relies solely on global PageRank (PR) centrality [9]. Our approach instead combines PageRank [10] with Personalized PageRank (PPR) [11] to build automatically extracted, semantically focused modules, and studies the impact of different verbalization strategies on the quality of the generated alignments. We implement this pipeline in VERSO (**VER**balization and **Str**uctural **mO**dularization), a schema-only complex matcher that requires neither instance data nor task-specific training.

The remainder of the paper is organized as follows. Section 2 presents related work. Section 3 details the proposed approach. Section 4 is devoted to the experimental evaluation. Section 5 reports the obtained results and analyses the performance of VERSO in comparison with state-of-the-art approaches. Section 6 discusses these results in light of the research questions. Finally, Section 7 concludes the paper and outlines perspectives for future work.

2. Related Work

Research in ontology matching has progressed thanks to effective techniques for simple alignments, based on lexical similarity, structural similarity, and machine learning. However, the identification of complex correspondences remains the most challenging subfield, requiring more advanced strategies.

Classical approaches to complex matching. CANARD [3] is a system guided by *Competency Questions for Alignment* (CQA), submitted as SPARQL queries based on the source ontology. The correspondences are generated by projecting the subgraph of the source CQA onto lexically similar neighborhoods of the corresponding instances in the target ontology. Evaluations show that CANARD favors coverage at the expense of precision, while being one of the few systems capable of handling large-scale knowledge graphs. Its performance nevertheless remains strongly dependent on data density and on the quality of *sameAs* links between instances. AROA [4] automatically generates simple and complex alignments between ontologies sharing common instances. The system extracts triples to build a typed transaction database, on which the Frequent Pattern Growth (FP-growth) algorithm mines association rules. These rules are then filtered through a catalogue of ten predefined simple and complex correspondence patterns, ranging from class equivalence to *Class by Attribute Type* and *Typed Property Chain* equivalence and subsumption. Although this catalogue covers m:n constructions, it fixes the admissible form of the correspondences a priori, so that any heterogeneity not anticipated at design time remains out of reach. Moreover, the ontologies to be aligned are required to share common instance data, since the mining step operates on transactions derived from individuals. Early work, Ritze et al. [12, 13] propose one of the first pattern-based approaches for detecting complex alignments, relying on *matching conditions* that instantiate atomic correspondence patterns defined by Scharffe [14] (e.g., *Class by Attribute Type*, *Property Chain*, *Inverse Property*) by combining lexical similarity between entity labels, structural constraints on property domains and ranges, and linguistic analysis. Similarly, AMLC [15], the complex alignment extension of AgreementMakerLight (AML) [16], exploits lexical similarity and structural conditions to detect the *Class by Attribute Type* and *Class by Attribute Existence* patterns. Both approaches, however, remain restricted to predefined patterns, which fixes the structure of the correspondences a priori and limits the expressiveness and coverage of the generated alignments.

Approaches based on language models. Matcha [17] extends the architecture of AML [16] with a core framework explicitly tailored to complex and holistic matching, and therefore addresses the same task as the present work. Among its matching algorithms, one relies on a pre-trained *sentence-BERT* model [18], used without fine-tuning, which represents the labels and synonyms of entities as embeddings subsequently compared through cosine similarity, allowing the capture of a semantic context beyond the explicit declarations of the ontology. The scope of this language model component, however, is confined to isolated lexical entries: it does not represent the axioms and neighbouring entities that define an entity within the ontology graph, and consequently cannot, on its own, assemble the logical constructors that complex correspondences require. The expressive correspondences produced by the system stem instead from the pattern-based algorithms inherited from AML and AMLC discussed above, and are thus subject to the same restriction to a predefined catalogue of patterns. Early explorations based on *Transformer* architectures highlighted their potential. Although originally designed for entity matching rather than ontology matching, the DITTO system [19] stands out as one of the first approaches to exploit models based on *Bidirectional Encoder Representations from Transformers* (BERT) [20] within the broader matching problem, demonstrating the potential of pre-trained language models for identifying correspondences between data entities. In the ontology matching setting, *BERTMap* [5] integrates the fine-tuning of *Transformer* encoders into matching tasks, combining it with alignment repair mechanisms to ensure the consistency of the resulting alignments. These systems target simple correspondences and derive their decisions from the similarity of isolated entity labels rather than from the structural context in which those entities are defined. Furthermore, *BERTMap* requires a fine-tuning phase over label sub-corpora extracted from the ontologies, whereas the approach proposed here operates without any task-specific training.

Subsequent works explore the use of generative LLMs in *zero-shot* and *few-shot* modes. The authors of [6] notably apply ChatGPT¹ to a small pair of ontologies (*Conference* track of OAEI), obtaining acceptable recall but insufficient precision, with an F1-score lower than that of a *baseline system* based on simple string similarities. This result illustrates the tendency of unconstrained LLMs to generate

¹<https://openai.com/index/chatgpt/>

System	No instance data req.	Automatic context	Targeted context	Representation given to model
CANARD [3]	✗	–	–	–
AROA [4]	✗	–	–	–
Matcha [17]	✓*	✗	✗	labels, syn.
BERTMap [5]	✓	✗	✗	labels
Amini et al. [8]	✓	✗	✗	formal syntax
CMatch [9]	✓	✓	✗	formal syntax
Ours	✓	✓	✓	NL triple

Table 1

Positioning with respect to closer work. *Targeted context* denotes whether the extracted context is adapted to the specific entity being aligned. “–” denotes dimensions that do not apply, as the system does not build a structured context around the entities to be matched. *Matcha includes an instance-based class matcher among its algorithms, but does not require populated ontologies to operate.

an excess of false positives [6, 21]. More recent works show that LLMs, when properly guided, can achieve remarkable performance. The authors of [22] thus demonstrate that GPT-3.5 [23] and GPT-4 [24], combined with embedding-based retrieval strategies, outperform the state of the art in biomedical ontology alignment. Likewise, LLMs4OM [25] shows that GPT-3.5 and LLaMA-2 [26] can match, or even surpass, traditional systems on several OAEI tracks thanks to *retrieval-augmented prompting* [27]. OLaLa [21] showed that labels and simple class descriptions alone already yield promising results under *zero-* and *few-shot* prompting, while [28] and GenOM [29] report that enriching ontologies with textual descriptions improves alignment coverage and precision.

From the perspective of combining LLMs with rigorous prompt design and effective retrieval strategies, [8] exploits the MOMo (*Modular Ontology Modeling*) methodology [30], which integrates ontology modules to provide a structured semantic context. The procedure consists in loading the source ontology as the initial *prompt*, then introducing the target entities to verify the alignments, enriching the *prompt* with *few-shot* descriptions, *Chain-of-Thought* strategies, and central axioms of the modules. This approach reaches good levels of precision and surpasses direct *prompting* attempts without modularization. On the other hand, it presents two main limitations: the modularization remains manual, and loading entire ontologies into the *prompts* entails a higher processing time. CMatch [9] improves upon the approach of [8] by introducing automatic modularization through the PageRank algorithm [10], thereby improving the scalability of the alignment process. CMatch relies on a *few-shot* strategy and selects, by cosine similarity, the k most relevant target modules as candidates for generation. Both approaches, however, provide the extracted modules to the model in formal syntax, and CMatch selects context on the basis of global importance alone.

Positioning of the proposed approach Table 1 summarises the discussion above along four dimensions that we consider critical for complex ontology matching: independence from instance data, automatic extraction of context, adaptation of that context to the entity being aligned, and the representation in which that context reaches the model. No existing system satisfies all four simultaneously. Instance-based and rule-mining systems achieve expressive correspondences at the cost of strong requirements on their input; systems built on pre-trained encoders operate at the schema level but reason over isolated lexical entries, delegating expressiveness to predefined patterns; and LLM-based approaches build context either manually or through global centrality, and deliver it in a formal syntax that does not correspond to the natural language distribution these models are predominantly trained on. The present work addresses this gap with a fully automatic, schema-only pipeline that enriches modularization by parametrizing module extraction through entity-specific relevance, via *Personalized PageRank* (PPR) [11], aiming at providing a more focused context and to capture more relevant entities for *matching*. Module content is delivered to the model as an explicit textual representation, through a triple verbalization module inspired by [31], fostering a better semantic understanding of the ontologies.

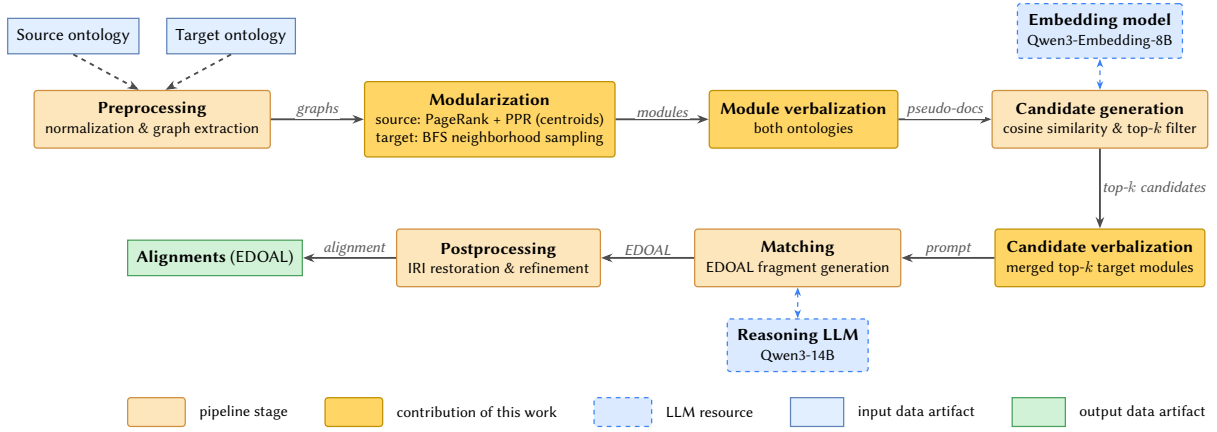


Figure 1: Overview of the proposed pipeline for complex ontology matching. Solid arrows denote the data flow between pipeline stages (labelled with the artefact they carry); dashed blue connectors denote the invocation of LLM resources, which are not pipeline stages: the candidate generation stage invokes the embedding model to embed the verbalized pseudo-documents, and the matching stage invokes the reasoning LLM with a few-shot prompt built from the verbalized candidate modules, receiving EDOAL fragments in return.

3. Proposed Approach

According to [32], the majority of matching systems follow a similar pipeline composed of three steps: preprocessing, candidate generation, and post-processing. The pipeline proposed here, hereafter referred to as *our pipeline*, enriches this architecture with two additional components: modularization and verbalization. Figure 1 presents an overview of the architecture, each component of which is detailed below.

3.1. Preprocessing Component

During preprocessing, the ontologies are loaded and then cleaned to refine the data used in the matching. The operations performed are: removal of instances (since the matching is carried out only at the schema level of the ontologies), removal of blank nodes to simplify the representation of the ontology, and removal of *disjointWith* relations. The last operation is motivated by the modularization step: *disjointWith* relations may harm the performance of PageRank by leading to the construction of modules containing semantically disjoint entities. For example, if *A* is declared *disjointWith* *B*, these two entities should not coexist within the same module. In some ontologies, entities can be associated with a large number of *disjointWith* relations, which may lead PageRank to assign them excessive importance. However, according to the definition of a module proposed by [30], only entities semantically related to the central entity of the module are relevant.

3.2. Modularization Component

The modularization step is inspired by the works of [8] and [1]. The former exploits ontology modules [30] to generate complex alignments using LLMs, under the assumption that modularization facilitates their generation. However, their modules are defined manually, and their approach requires loading the entire target ontology into the prompt in order to identify alignments from the modules, which limits both its scalability and its applicability to large ontologies. Building upon this notion of ontology modularization, [1] operationalize module extraction through a topological heuristic based on the PageRank algorithm [10]. In their approach, the entities with the highest PageRank scores are identified as centroids around which modules are constructed by means of a Breadth-First Search (BFS) traversal that selects the closest neighbors of each centroid up to a depth of 1. This heuristic-driven formulation removes the need for manual module definition, reduces the search space to a set of semantically central entities, and thereby improves the scalability of modularization-based matching.

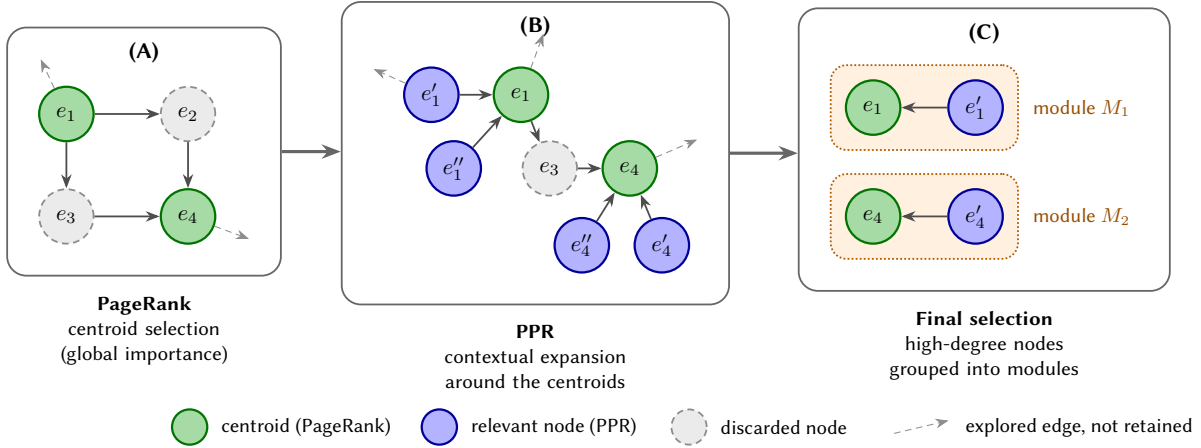


Figure 2: Overview of the modularization component. Flow (A) corresponds to the selection of centroids by *PageRank* (in green), reflecting the global importance of nodes in the source ontology. Flow (B) represents the contextual expansion performed by *PPR* around each centroid, capturing the most relevant nodes (in blue); discarded nodes and explored but non-retained edges are shown with dashed strokes. Flow (C) corresponds to the final selection of high-degree nodes: each centroid and its retained neighbors are grouped into a module (orange boxes) used for the modularization of the source ontology.

In our approach, PageRank is likewise employed as a *global* importance heuristic: concepts with high PageRank scores tend to be connected to many other concepts, or to concepts that are themselves central, thus acting as semantic hubs within the ontology. These global centrality values can guide the alignment process by highlighting potentially more relevant or more influential entities. For example, classes that are highly central in both ontologies can serve as starting points for the search for correspondences, and PageRank scores can be exploited as weights when combining similarity measures, assigning greater importance to correspondences involving central concepts. The novelty of our modularization strategy with respect to [1] is twofold. First, we remove *disjointWith* relations prior to computing centrality, preventing PageRank from assigning excessive importance to entities that are heavily involved in disjointness axioms and from grouping semantically disjoint entities within the same module. Second, and in contrast to plain PageRank, we additionally employ Personalized PageRank (PPR) [11] to refine the construction of *source* modules. Whereas PageRank answers the question “how important is node x within the entire graph?”, PPR answers “how relevant is node x with respect to a given node y ?”, by biasing random walks toward a set of seed nodes. In our pipeline, the centroids identified by PageRank serve as seeds for PPR, so that the resulting modules are populated with the neighbors that contribute most strongly to the semantic definition of the entity under consideration, rather than with overly generic concepts. The two algorithms are therefore complementary rather than interchangeable: PageRank selects *which* entities anchor the modules, while PPR determines *what contextual content* each module should contain.

3.3. Verbalization Component

The verbalization component translates the modules extracted in the previous stage into textual descriptions, hereafter *pseudo-documents*, consumable by LLMs. Serializations such as *Turtle*, used in [1], are machine-processable but do not constitute a textual input that LLMs exploit effectively [33]. The component is built on top of *OntoAligner* [34], employed for two complementary purposes: the extraction of RDF triples $\langle s, p, o \rangle$ from each module, and the retrieval of the lexical annotations (e.g., `rdfs:label`) attached to each entity. Given a module M extracted around a centroid entity, the triples of M are verbalized and concatenated into its pseudo-document. In the *natural* strategy, predicates are mapped to linguistic patterns following the template-based approach of DeepOnto [35], whereby `rdfs:subClassOf` is rendered as “ A is a type of B ”; the remaining strategies preserve the predicate as in the triple.

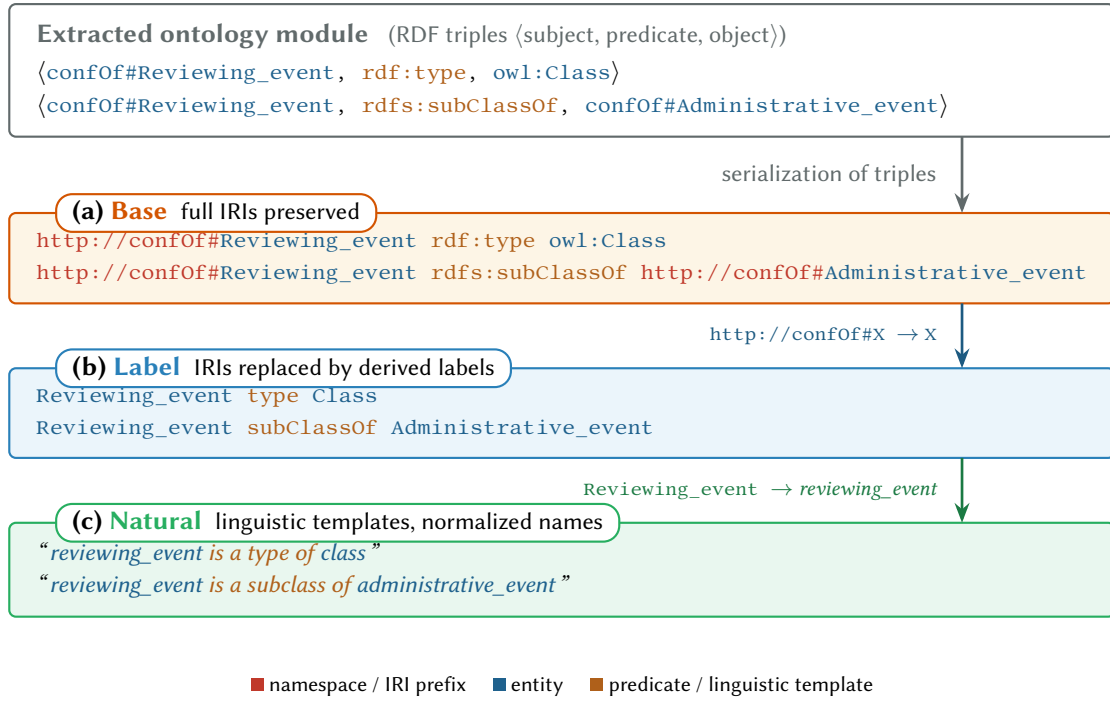


Figure 3: The three verbalization strategies applied to an extracted ontology module. The *base* strategy (a) serializes the triples while preserving full IRIs; the *label* strategy (b) replaces IRIs with the labels derived from them; the *natural* strategy (c) reformulates the triples as natural language sentences via predefined linguistic templates, normalizing entity names to lowercase while preserving the underscore between their constituent terms. Color coding distinguishes namespace prefixes, entities, and predicates across the three levels of abstraction.

Three strategies of increasing linguistic abstraction are compared (Figure 3). **Base** (a) preserves the full *Internationalized Resource Identifiers* (IRIs) of each entity, retaining namespace prefixes at the cost of readability. **Label** (b) substitutes the IRIs by the corresponding labels, removing namespace noise while keeping the original identifier morphology (underscores, CamelCase). **Natural** (c) further normalizes compound names to lowercase, separates their constituent terms by spaces, and applies the predicate templates, yielding the text closest to the pre-training distribution of language models.

Verbalization is invoked at two distinct points of the pipeline (Figure 1): on the modules produced by the modularization component, whose pseudo-documents feed embedding generation and cosine similarity computation; and on the top- k candidate modules retrieved from the target ontology, which are merged into a single module and verbalized anew before being injected as context into the few-shot prompt of the alignment LLM. The two label-to-IRI dictionaries (one per ontology) built at this stage are reused by the post-processing component to restore the original IRIs in the generated *Expressive and Declarative Ontology Alignment Language* (EDOAL) fragments (Section 3.6).

3.4. Candidate Generation Component

This component relies on an LLM model to generate embeddings of the modules from the source and target ontologies, the similarities of which are computed via cosine similarity, leading to the construction of a similarity matrix. For each centroid of a module of the source ontology identified by PageRank, a set of nodes of the target ontology is selected and then verbalized. The k target modules most similar to the source module, according to cosine similarity, are retained as candidates for alignment. The candidate modules thus retrieved are then provided as context to the language model, which generates the corresponding alignments. The parameter k is a hyperparameter controlling the trade-off between candidate coverage and computational complexity.

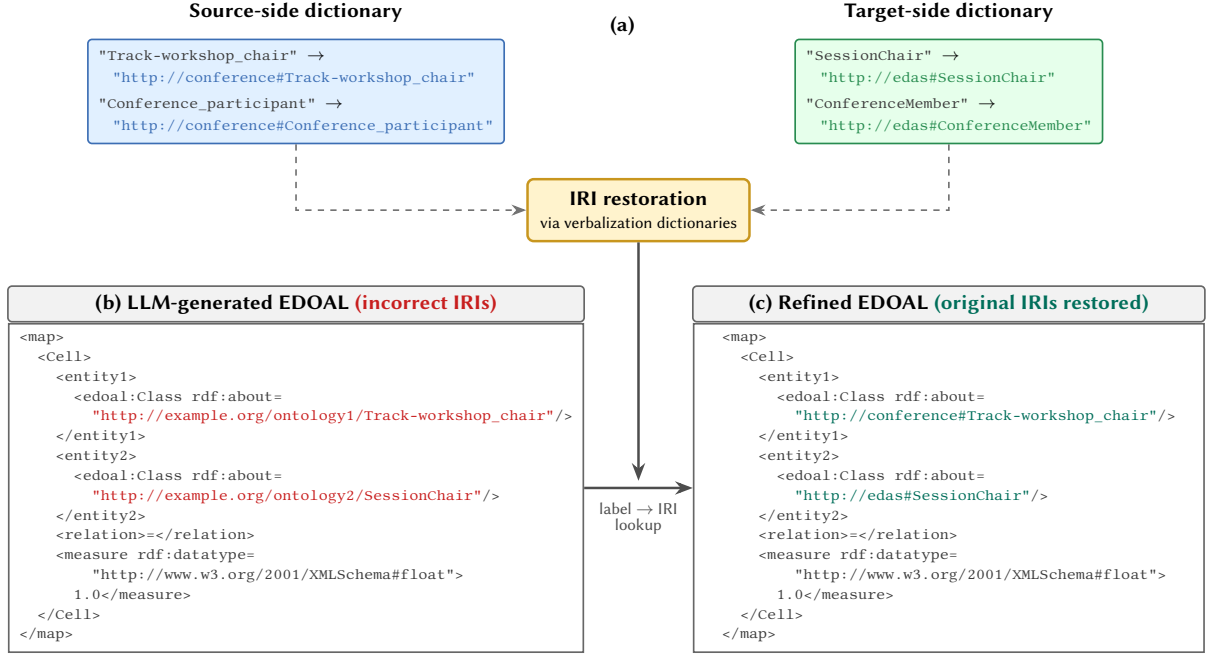


Figure 4: Overview of the IRI restoration mechanism of the post-processing component. (a) During verbalization, two label-to-IRI dictionaries (one per ontology) are generated. (b) The EDOAL fragments produced by the LLM reuse the generic `http://example.org/` prefix inherited from the few-shot examples of the prompt (in red). (c) Each label is looked up in the corresponding dictionary and replaced by its original IRI, yielding a refined EDOAL document evaluable against the reference alignments.

3.5. Alignment Component

The alignment component relies on a reasoning LLM, capable of decomposing complex problems into intermediate steps, in charge of generating the alignments from the information provided in the *prompt* in EDOAL format. A *few-shot* strategy is adopted, providing a limited number of examples to guide the model. The *prompting* strategy follows that of [1], where two examples of pairs of ontology modules are provided in *Turtle* format, adapted to the type of verbalization employed. The general structure of the *prompt* follows the work of [9].

3.6. Post-Processing Component

The post-processing component aggregates the LLM outputs in EDOAL format into a single document containing the set of generated correspondences, then applies a refinement phase to correct the errors introduced by the model. In the configurations using the *label* and *natural* verbalizations, the LLM frequently generated entities with IRIs reusing the generic prefix `http://example.org/ontology1/`, identical to the one used in the *few-shot* examples of the prompt. To address this problem, two dictionaries associating labels with the original IRIs are generated during the verbalization step, one for each ontology. After generation of the EDOAL document, the labels produced by the LLM are replaced by the corresponding IRIs using these dictionaries. Figure 4 illustrates this refinement process.

4. Experimental Evaluation

This section presents the experiments conducted to address the following research questions:

- *RQ1: To what extent can the use of ontology modularization techniques improve the performance of the matching system?*
- *RQ2: To what extent do different verbalization strategies affect alignment performance, as measured by the F1-score of the generated correspondences against the reference alignments?*

The remainder of this section is organized as follows: Section 4.1 describes the datasets employed in the experiments and their main statistics, while Section 4.2 details the experimental settings.

4.1. Datasets

Four OAEI datasets from the *Complex* track are used: *Conference*, *Hydrography*, *GeoLink*, and *Taxon*². In total, 15 ontologies were considered, corresponding to 17 pairs of ontologies³:

- **Conference** (10 pairs): *cmt-conference*, *cmt-confOf*, *cmt-edas*, *cmt-ekaw*, *conference-confOf*, *conference-edas*, *conference-ekaw*, *confOf-edas*, *confOf-ekaw*, *edas-ekaw*;
- **GeoLink** (1 pair): *gbo-gmo*;
- **Hydrography** (3 pairs): *hydro3-swo*, *hydrOntology-swo*, *cree-swo*;
- **Taxon** (3 pairs): *taxon-agrovoc*, *taxon-dbpedia*, *taxon-taxref*.

Table 2 presents the statistics relating to the number of ontologies, triples, and simple and complex correspondences for each of the considered datasets.

Table 2
Summary of the datasets used

Dataset	Conference	Geo	Hydro	Taxon
Ontologies	5	2	4	4
Triples	3867	2720	17196	776690
Simple Correspondences	111	19	113	6
Complex Correspondences	184	48	84	20

4.2. Experimental Settings

Preprocessing component. The ontologies were loaded using the *rdflib*⁴ and *networkx*⁵ libraries. The ontology is first loaded into *rdflib*, then translated into a *networkx* graph representation. This translation step is motivated by the fact that *networkx* provides a set of algorithms particularly useful for graph processing and analysis.

Modularization component. The PageRank (PR) and Personalized PageRank (PPR) algorithms were configured with a damping factor of 0.8 and 100 computation iterations.

Verbalization component. The verbalization component relies on OntoAligner [34], which generates triples from the ontologies as well as the labels associated with the ontology entities.

Candidate generation component. The parameters of the candidate generation component are identical to those of CMatch. The embedding model used is Qwen3-Embedding-8B⁶, with a batch size of 2 and a maximum context length of 8192 tokens.

Alignment component. The Qwen3-14B model⁷ is used for the alignment phase (matcher), with a maximum context length of 8192 tokens, a temperature of 1.0, and a fixed seed guaranteeing the reproducibility of the outputs.

²<https://oaei.ontologymatching.org/2025/complex/index.html>

³<https://github.com/liseda-lab/complex-OM-benchmark/tree/main>

⁴<https://rdflib.readthedocs.io/en/stable/>

⁵<https://networkx.org/en/>

⁶<https://huggingface.co/Qwen/Qwen3-Embedding-8B>

⁷<https://huggingface.co/Qwen/Qwen3-14B>

Language models. The embedding models were selected on the basis of the *Massive Text Embedding Benchmark* (MTEB) [36]. The Qwen3-Embedding-8B model was retained, ranking among the top three open-source models according to this *benchmark*. The reasoning model used for the alignment task is Qwen3-14B, loaded via the *HuggingFace* API⁸.

Evaluation metrics. The evaluation metrics are those adopted by the OAEI for the considered datasets, in particular the *Tree Edit Distance* (TED) [37].

The experiments presented in the next section were evaluated based on the following possible combination of settings: *Topology*: with or without PPR; *Neighborhood depth*: 1 and 2; *Number of candidates generated*: 2, 3, and 5; and *Verbalization*: base, label, and natural.

5. Results and Discussion

The results in this section are presented as follows: for each dataset, the configuration achieving the best F1-score is first identified, and its counterpart with PPR modularization is then examined (or, if PPR is already part of the best configuration, its counterpart without PPR is examined instead).

Figure 5 presents the results on the *Conference* dataset, compared to the CMatch and CANARD systems. The best-performing configuration combines the *base* verbalization, a BFS exploration of depth 1, without PPR, surpassing CANARD and CMatch on this dataset. On *Conference*, PPR increases recall (0.362 to 0.438) but reduces precision (0.334 to 0.238). We hypothesize that the drop in precision is due to the *Lost in the Middle* phenomenon [38] because the increase in recall makes the context window longer and confuses the model.

Figure 6 presents the results on the *Hydrography* dataset, compared to the performances of the CMatch and Matcha systems. VERSO outperforms both baselines by 15 to 20 percentage points, achieved with the *base* verbalization, a BFS exploration of depth 2, and without PPR. The same hypothesis about the drop of precision for PPR approach in the previous experiment on the *Conference* dataset falls here too.

Figure 7 presents the results on the *GeoLink* dataset, compared to the performances of the CANARD, CMatch, and AROA systems. Under the same configuration, that is, *base* verbalization, BFS of depth 2, and no PPR, VERSO exceeds CMatch by 3 to 4 percentage points, but remains behind CANARD and AROA. GeoLink is the only dataset on which PPR achieves higher recall than the plain PageRank approach, although the improvement is marginal and not statistically significant.

Figure 8 presents the results on the *Taxon* dataset, compared to the performances of the CMatch and CANARD systems. Our system surpasses CMatch and improves the performance of CANARD by approximately two percentage points, with the *base* verbalization, a neighborhood depth of 1, and without PPR. The same reason for the first 2 datasets falls here too.

Overall performance across datasets In Figure 9, the average of the results over the entire set of datasets (*Conference*, *Hydrography*, *GeoLink*, and *Taxon*) is compared between the systems. Our approach exceeds CMatch on average by more than 20 percentage points, with the *base* verbalization, a BFS exploration of depth 1, and without PPR. Figure 10 presents the average of the results on the *Conference*, *GeoLink*, and *Taxon* datasets, compared to the performances of the CANARD system. On this three-dataset aggregate, CANARD obtains a higher overall F1-score (0.329) than the best configuration of VERSO (0.293), driven primarily by its performance on GeoLink. It should be emphasized that CANARD requires the presence of instances to perform the matching. Conversely, VERSO relies solely on the schema information of the ontologies, without resorting to instances, which gives it a notable advantage in terms of genericity and applicability, imposing fewer prerequisites on the input ontologies. Furthermore, VERSO obtains better F1-scores on the *Conference* and *Taxon* datasets when they are considered individually.

⁸<https://huggingface.co/models>

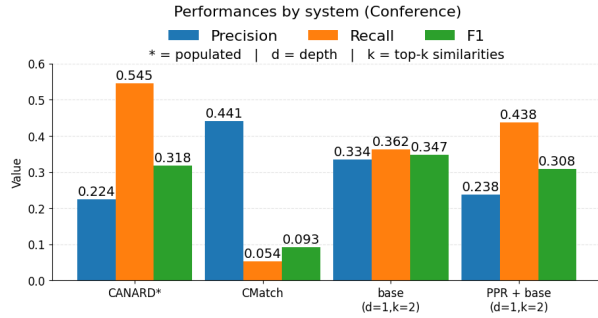


Figure 5: Results on the *Conference* dataset
(* = populated)

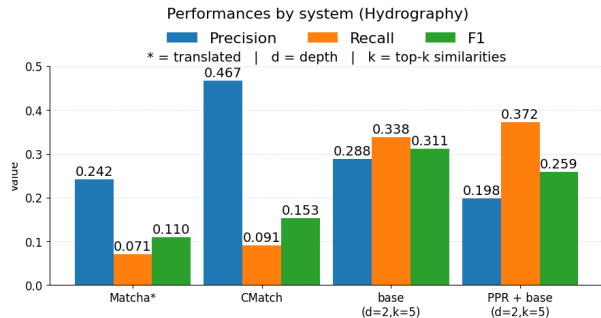


Figure 6: Results on the *Hydrography* dataset
(* = translated)

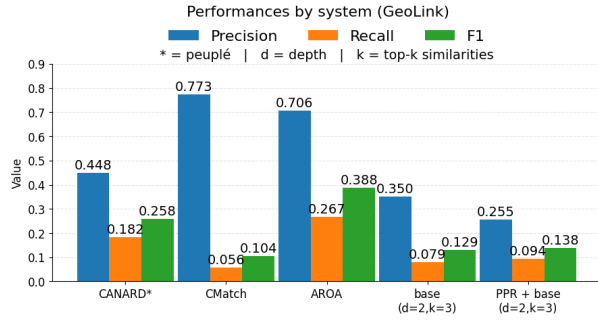


Figure 7: Results on the *GeoLink* dataset.

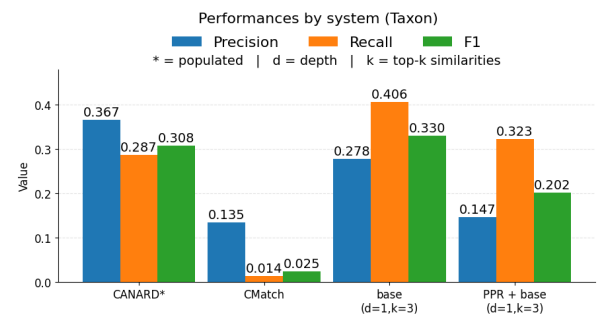


Figure 8: Results on the *Taxon* dataset.

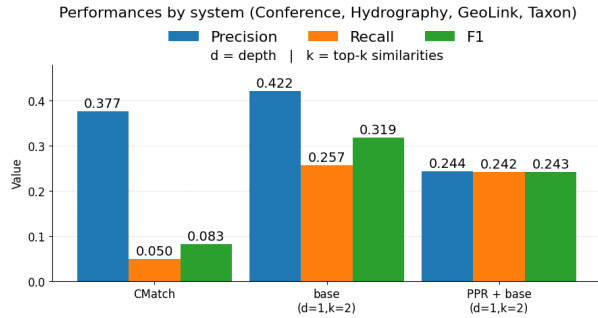


Figure 9: Average performances with CMatch.

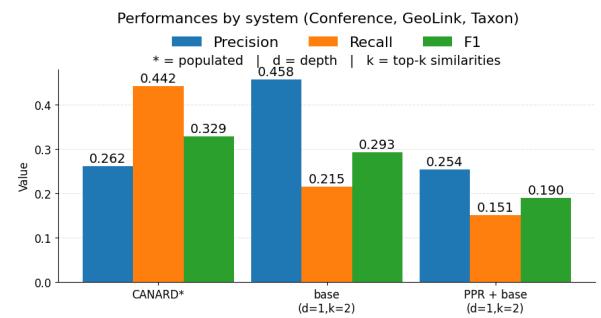


Figure 10: Average performances with CANARD.

In response to **RQ1**, concerning modularization, the integration of PPR produces a lower F1-score than the configuration without PPR, probably due to the increased volume of information submitted to the LLM, which degrades the quality of the alignments. Indeed, language models tend to lose focus on the central information when faced with excessively long or dense *prompts* (the *Lost in the Middle* phenomenon [38]). In response to **RQ2**, concerning verbalization, the *base* configuration produces better results because the LLM tends to faithfully reproduce the structure provided in the examples of the *prompt*. The results show that the *base* verbalization offers the best average performance, in combination with a BFS exploration of depth 1, and without PPR. This aspect is decisive, since the evaluation rests on a strict comparison between the alignments produced in EDOAL format and the reference alignments, which requires an exact correspondence of the aligned entities.

6. Conclusion and Future Work

This paper introduced an LLM-based approach for complex ontology matching that combines topological modularization (using PageRank and Personalized PageRank to extract entity-centered modules) with a

verbalization module that converts these modules into text. Evaluated on four OAEI Complex track datasets, the approach generates expressive correspondences without requiring populated ontologies and outperforms comparable LLM-based systems. Additionally, biasing module extraction toward entity-specific relevance reduced precision across all datasets while improving F1-score in only one case, and the least abstract verbalization strategy (preserving entity IRIs) achieved the best F1-score in three of the four datasets.

Despite these results, the main limitation of this work is the lack of experimental evidence for some of its hypotheses, including the possible link between the Lost in the Middle phenomenon and the precision drop in PPR-based modularization, and the claim that IRI-specific tokens inflated similarity scores for the base verbalization strategy. Another limitation concerns PPR itself: being a graph-based algorithm, it considers only the structure of the graph, and not its semantics, to retrieve terms related to a given entity of the ontology. Building on these limitations, future work will focus on a quantitative and qualitative analysis of each component (PPR-based modularization and verbalization) to measure its contribution to the overall performance of the system. We also plan to investigate a hybrid modularization approach, combining a semantic dimension, based on embeddings, with a topological one, based on graph algorithms. Larger ontologies will also be evaluated to assess the scalability of the proposed approach.

Supplemental Material Statement: The source code required to reproduce the experiments presented in this paper is publicly available at <https://github.com/igorvlnascimento/VERSO>.

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Declaration on Generative AI

During the preparation of this work, the authors used Grammarly in order to: grammar and spelling check, paraphrase and reword. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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